Assay Reference Handbook

Orchid Cellworks | Synthetic study operations | Alpha A-24 and Beta B-16 | 2026-07-13

Purpose and boundary

This internal handbook standardizes how synthetic assay references are read, converted, and recorded. It applies only to Orchid Cellworks training cohorts Alpha A-24 and Beta B-16. It is not a clinical method, a patient record, or an external standard.

Reference workflow

Stage	Operator check	Record outcome
Prepare	Confirm study label, run date, and expected concentration basis.	Use the synthetic cohort label exactly as supplied.
Measure	Keep the observed reading and dilution factor as separate fields.	Do not embed dilution in the concentration unit.
Convert	Apply the unit relationship before rounding.	Record source value, converted value, and unit.
Review	Compare against the local reference band.	Flag a mismatch for internal review; do not overwrite observations.

Unit relationships

For aqueous concentration notation, 1 ng/mL equals 1 pg/uL because both numerator and denominator shift by 1,000. A conversion changes the expression, not the measured amount. Examples: 6.8 ng/mL equals 6.8 pg/uL; 1 ug/mL equals 1000 ng/mL; and 0.5 mg/mL equals 500 ug/mL.

Rounding and reporting

Preserve the source precision until the final reporting step. A displayed assay anchor of 6.8 ng/mL is reported as 6.8 pg/uL after the direct ng/mL-to-pg/uL conversion. Use a trailing zero only when it communicates deliberate precision from the source record.

Reference controls

Interpretation guide for synthetic training runs

Alpha A-24 / Beta B-16 comparison frame

Reference	Expected use	Example interpretation
Alpha A-24	Primary demonstration cohort	Use 6.8 ng/mL as the teaching anchor when the record specifies the Alpha reference.
Beta B-16	Secondary comparison cohort	Use the assigned Beta reference value; do not substitute the Alpha anchor.
Blank control	Background check	Record as blank; never represent it as a cohort value.
Conversion check	Unit-format check	Verify matching mass and volume shifts before export.

Escalation cues

Pause local entry when the instrument label differs from the run sheet, when the unit basis is absent, when an unknown dilution factor appears, or when a value is outside the stated synthetic reference band. Capture the discrepancy as a note and route it to the Orchid Cellworks internal review queue. Do not invent a missing source value.

Quick checklist

1. Confirm the synthetic cohort label. 2. Confirm the assay concentration unit. 3. Record dilution separately. 4. Convert with dimensional equivalence. 5. Retain the source reading and final unit. 6. Mark any unresolved mismatch for review.

Document status: synthetic internal training reference. No human specimens, clinical decisions, or external citations are represented.